Orallexa / Shadow Mode A Loan Origination Compliance Package

Complete technical report, design, implementation plan, and traceability package.

1. Executive Summary

This package resolves the gap between the BRD and the Shadow Mode A benchmark by preserving the BRD as the source for risk analytics and governance while using Addenda A-C as the source for loan-origination thresholds. It implements a complete analysis-only loan council with risk computation, policy checks, adverse-action mapping, audit guardrails, and tests.

2. Source Separation Principle

The BRD supports VaR/ES, the 10-day horizon, 95% confidence, Risk Core, audit controls, and no-trade governance. It does not explicitly support FICO 700, DTI 0.36, LTV 0.80, or VaR 0.12 as cutoffs. Those benchmark thresholds are supplied by Addendum A, Addendum B, and Addendum C.

3. Resulting Traceability Matrix

Benchmark Rule	Source	Implementation File	Test File
FICO >= 700	Addendum A	loan/policy.py	tests/test_loan_policy.py
DTI <= 0.36	Addendum B	loan/policy.py	tests/test_loan_policy.py
LTV <= 0.80	Addendum C	loan/policy.py	tests/test_loan_policy.py
VaR <= 0.12	Addendum C Risk Appetite Note	risk/risk_core.py	tests/test_risk_core.py
VaR/ES Framework	BRD Risk Core Specification	risk/var_es.py	tests/test_risk_core.py
10-Day Horizon	BRD Risk Packet Methodology	risk/models.py	tests/test_risk_core.py
Confidence 95%	BRD Risk Packet Methodology	risk/models.py	tests/test_risk_core.py
Analysis Only	BRD Governance Controls	risk/audit.py	tests/test_audit_and_council.py

4. Architecture

Loan Application -> Loan Schema Validation -> Addendum Policy Engine + BRD Risk Core -> Five Voice Council -> Verdict Resolver -> Analysis-Only Auditor -> Human Review Report.

5. Implementation Modules

risk/var_es.py: Historical simulation VaR and Expected Shortfall.

risk/risk_core.py: Builds risk packet at 95% confidence and 10-day horizon.

risk/audit.py: Blocks execution/trading language and enforces analysis-only output.

loan/policy.py: Applies FICO, DTI, and LTV thresholds from Addenda A-C.

loan/adverse_action.py: Maps policy failures into adverse-action reason codes.

agents/council.py: Runs the five-voice Mode A council and final resolver.

api/main.py: FastAPI endpoint for Shadow integration.

6. Test Plan

The test suite verifies threshold behavior, VaR/ES defaults, VaR cutoff handling, analysis-only governance, and final council traceability.

7. Runbook

Install dependencies with pip install -r requirements.txt, run pytest, execute scripts/run_demo.py, and optionally run the FastAPI route with uvicorn orallexa_modea.api.main:app --reload.

8. Conclusion

The package is benchmark-defensible because every threshold is tied to the correct source: BRD for the risk framework and governance, Addenda A-C for loan underwriting and deterministic cutoff values.